Spatial Statistics for Decision-Making



A Blackfly Tale

Dr. Claudio Fronterre

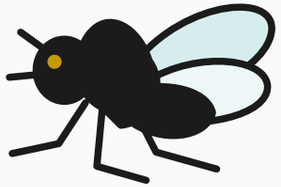
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The Future of Parasitology

12–13 October 2026

Onchocerciasis

Onchocerca volvulus causes skin disease and eye damage, including blindness.



Blackflies *(Simulium)*

Vectors breed in fast-flowing rivers

⇄

Transmission through blackfly bites



People

Infected people carry the parasite

> Repeated ivermectin treatment aims to interrupt transmission.

What evidence supports stopping treatment?

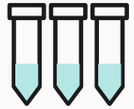
Evidence from people and vectors across the assessment area



Epidemiological assessment

Evidence of exposure to *O. volvulus* in children

Ov16 serology



Entomological assessment

Evidence of infective parasites in blackflies

O-150 PCR pool screening

Assess the evidence → Stop treatment when criteria are met → Post-treatment surveillance

Collecting enough blackflies for assessment

6,000

blackflies for entomological assessment across a transmission zone

Where should teams collect them?

Field time and resources are limited.
Choosing where to allocate effort is crucial.

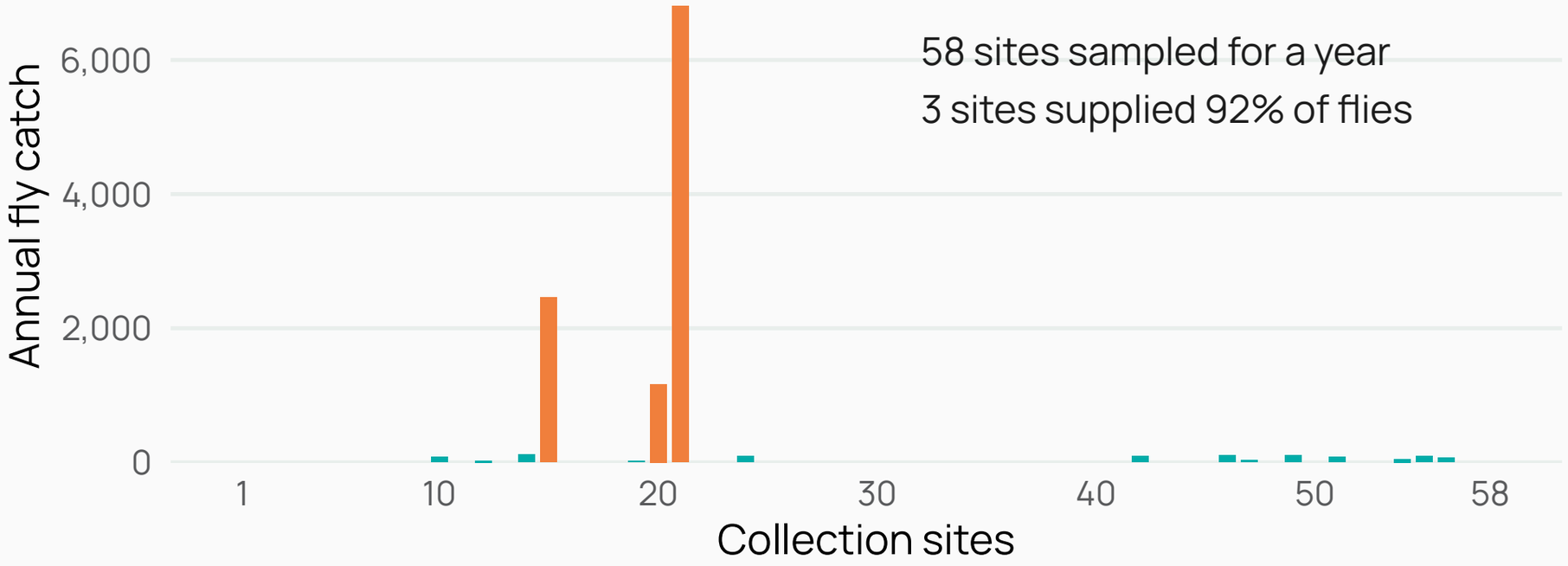
Entomological threshold: <0.05% infective flies

Upper 95% confidence bound · fewer than 1 in 2,000 flies
6,000 tested with no infective larvae detected provides the required precision.

WHO (2016). *Guidelines for stopping mass drug administration and verifying elimination of human onchocerciasis.*

The problem that started this work

Anambra · 2021–2022 entomological assessment



FMoH / The Carter Center

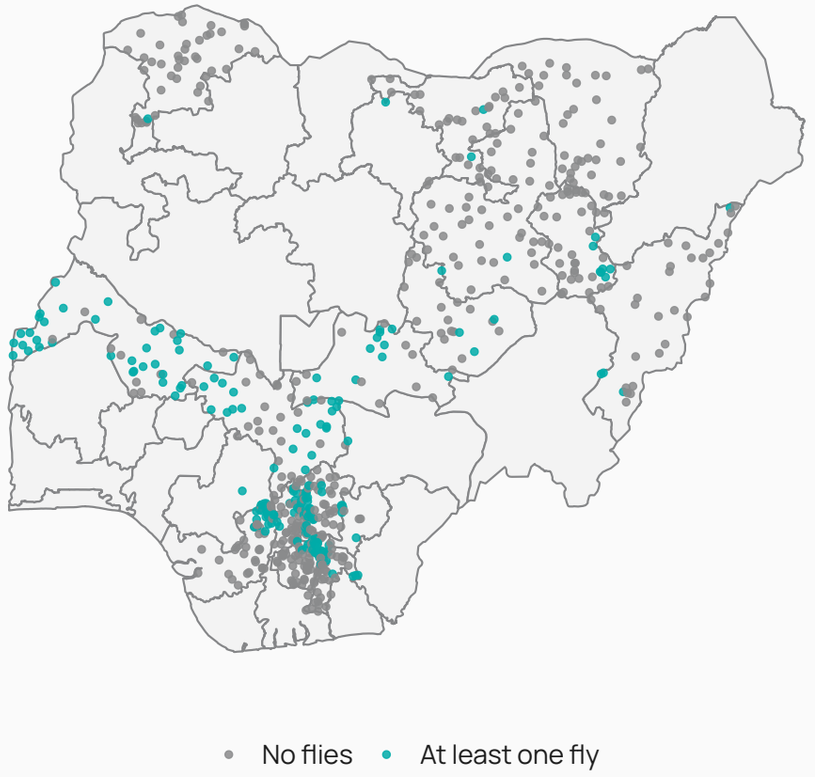
The Nigerian data behind Blackfly Finder

676 trap locations

7,617 observed site-months

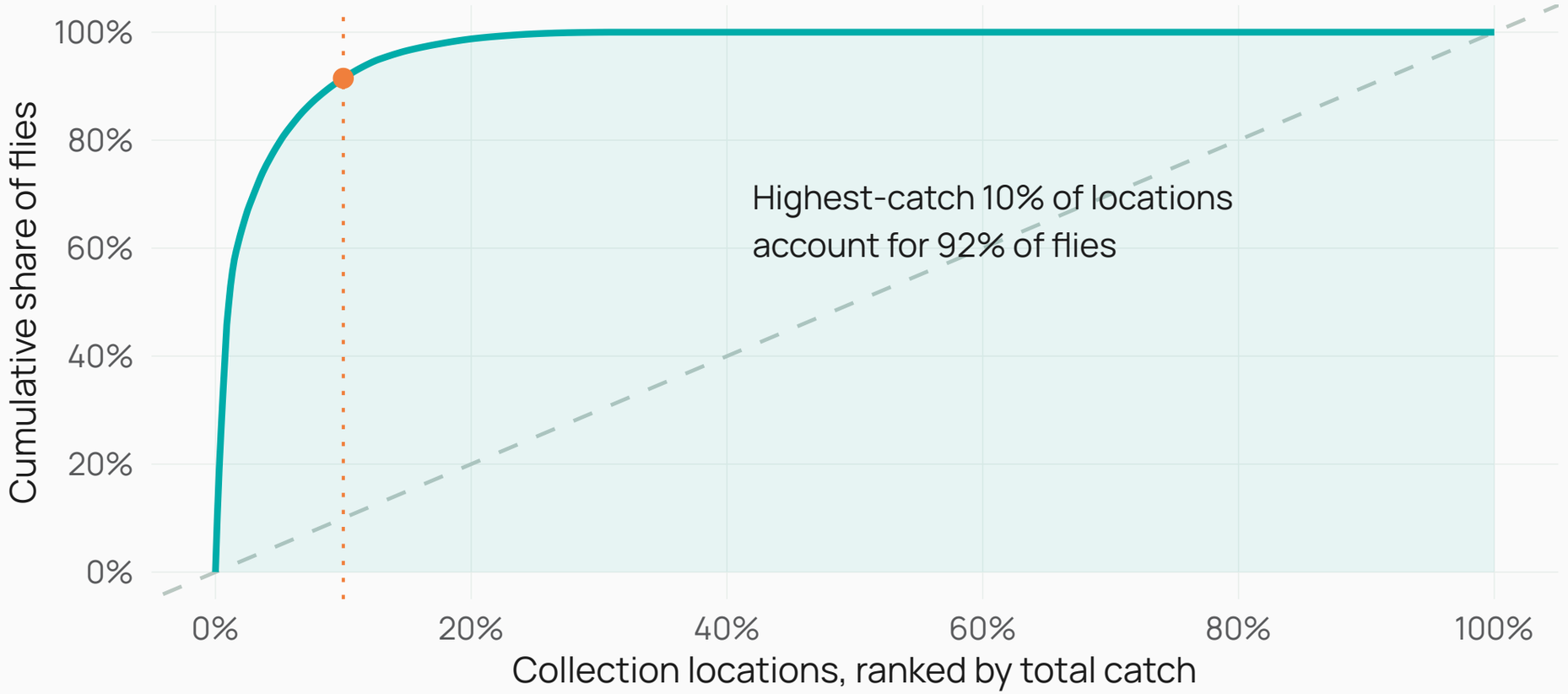
2020–2025 monthly collection records

Human landing collections



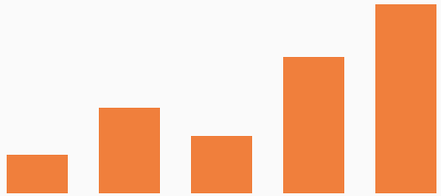
FMoH / TCC and SORT records

Catch is concentrated in a few sites



Blackfly Finder: a two-part hurdle model

0 / +



Will we find flies?

Probability of a positive catch

Ensemble of presence models

How many might we collect?

Expected catch when flies are found

Negative-binomial count model

Expected catch = Probability of finding flies × Expected positive catch

Allows for differences in observation duration.

What informs Blackfly Finder?

Rivers and terrain
Hydrology and landscape shape

Climate
Temperature and rainfall

Soils
Soil properties and temperature

Land and vegetation
Land cover, population and livestock

Presence / absence

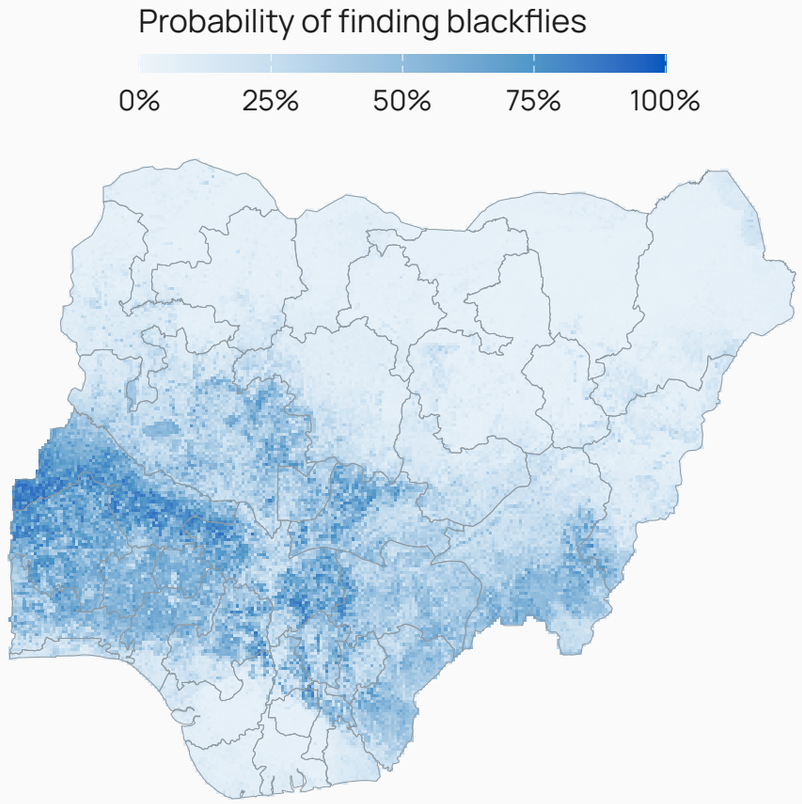
Environmental inputs combined in
an ensemble of presence models

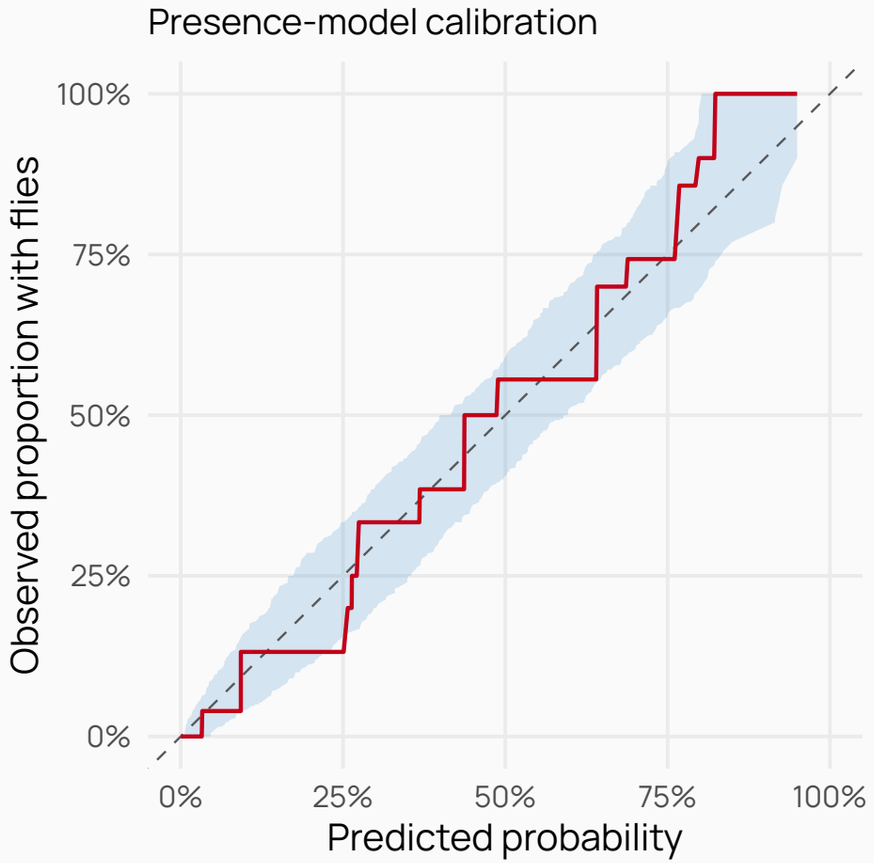
Positive-catch intensity

Air temperature range · soil clay content
Soil temperature seasonality · terrain variation

Different predictors can explain where flies are found and how many are caught.

Where should teams look for blackflies?





Cross-validated base predictions
Shaded 90% consistency band

Where can teams collect?

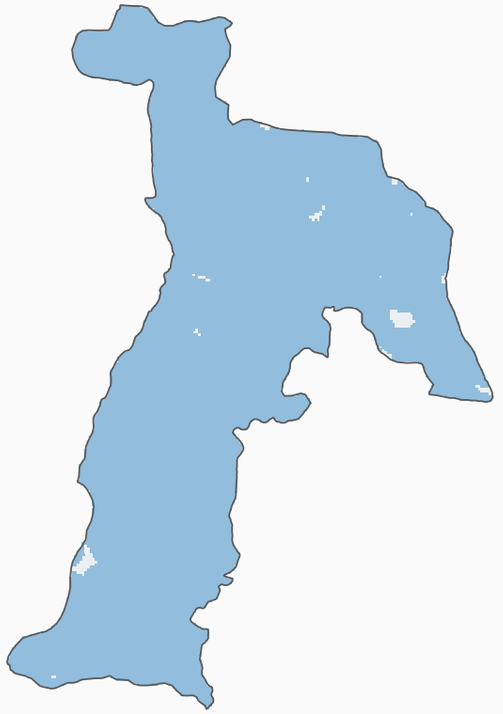
Define a feasible sampling region.

5 km from rivers

2 km from roads

5 km from populated areas

A location must satisfy all three access limits.



Abia · blue cells satisfy all three limits

Illustrative settings · teams can adjust the limits to local field conditions.

Choosing sites within the feasible region

Abia example · 30 proposed sites

Prioritise finding flies

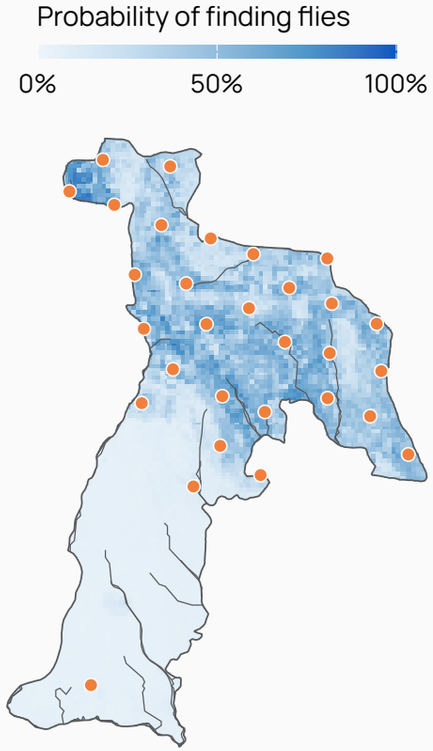
Rank feasible locations by their predicted probability.

Keep sites apart

At least 5 km between selected sites.

Check major-river coverage

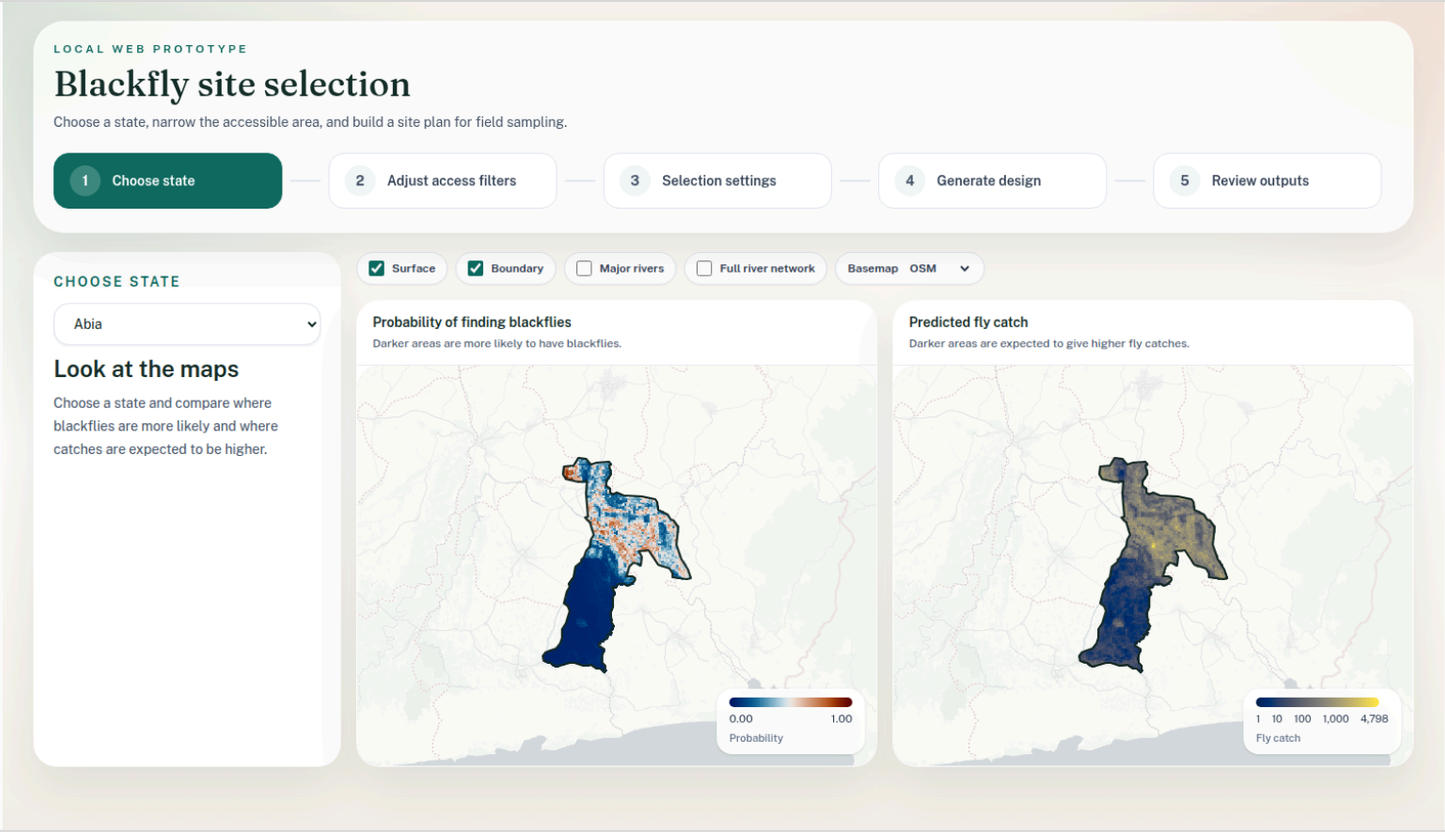
A selected site within 30 km of each sampled river point.



Orange: proposed sites · grey: major rivers

Proposed design for review · access, spacing and river coverage are checked together.

A sampling web-tool teams can use



1. Set constraints

Choose the state, access limits and site spacing.

2. Generate and review

Inspect the proposed sites and coverage.

3. Download the plan

Export coordinates and the design summary.

Could an adaptive strategy improve sampling?

Three monthly visits to the same site. No flies caught.

Month 1	Month 2	Month 3
0 flies	**0** flies	**0** flies

Would you stay at this site, or move?

Stay at the site

A more productive season may be ahead.

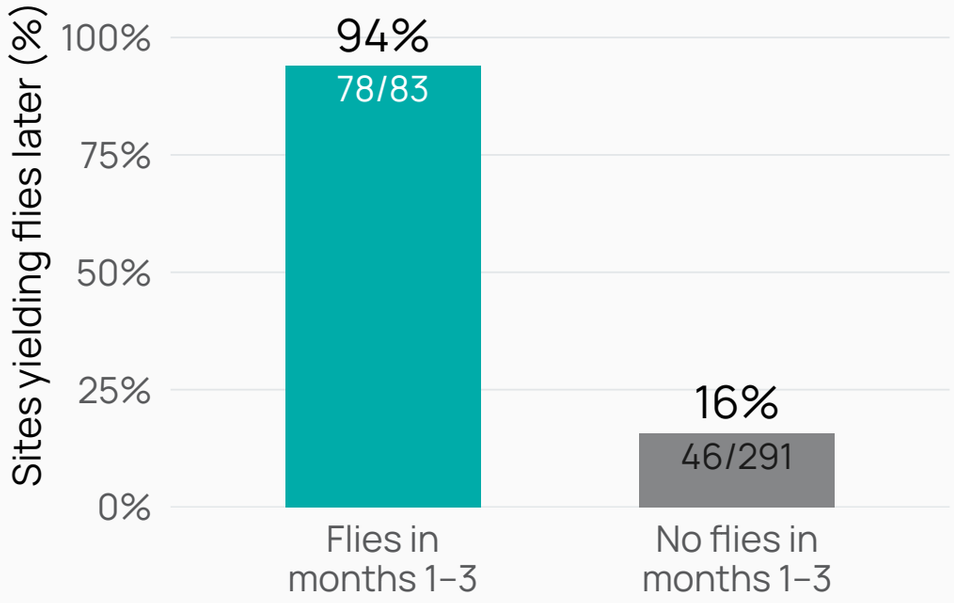
Move to a reserve site

Another site may offer a better yield.

What do repeated zeros imply?

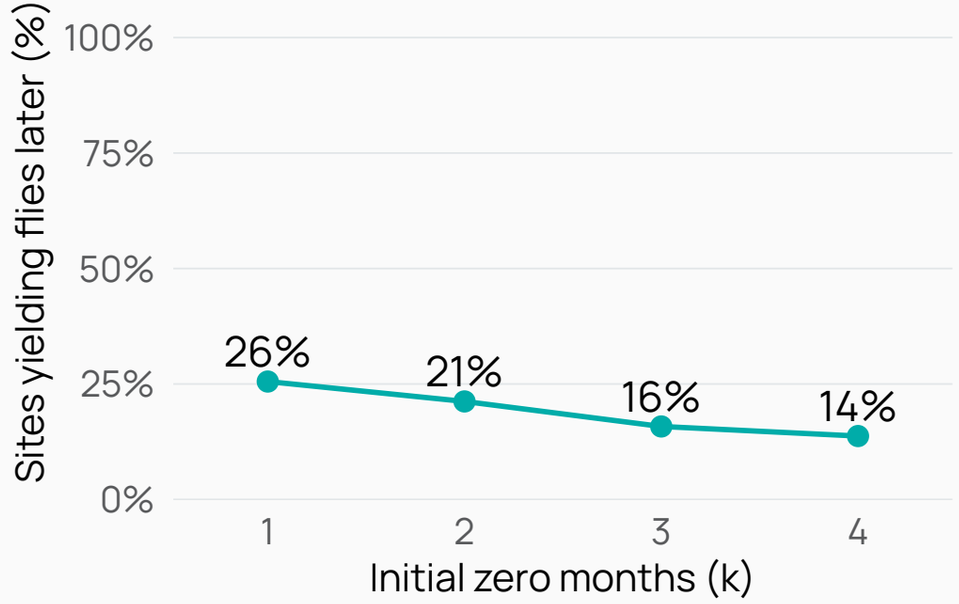
Early catches distinguish later success

Months 4–12



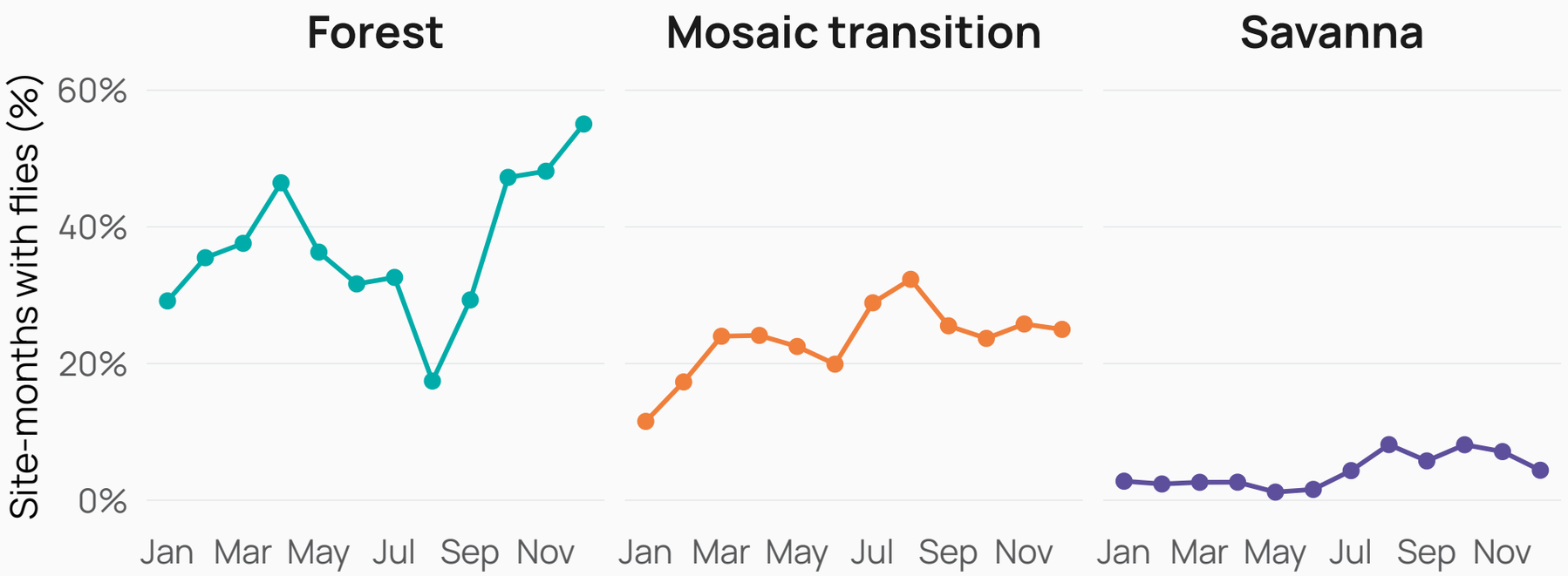
What follows a run of zeros?

Remaining 12 − k months



Early catches help identify sites more likely to yield flies later.

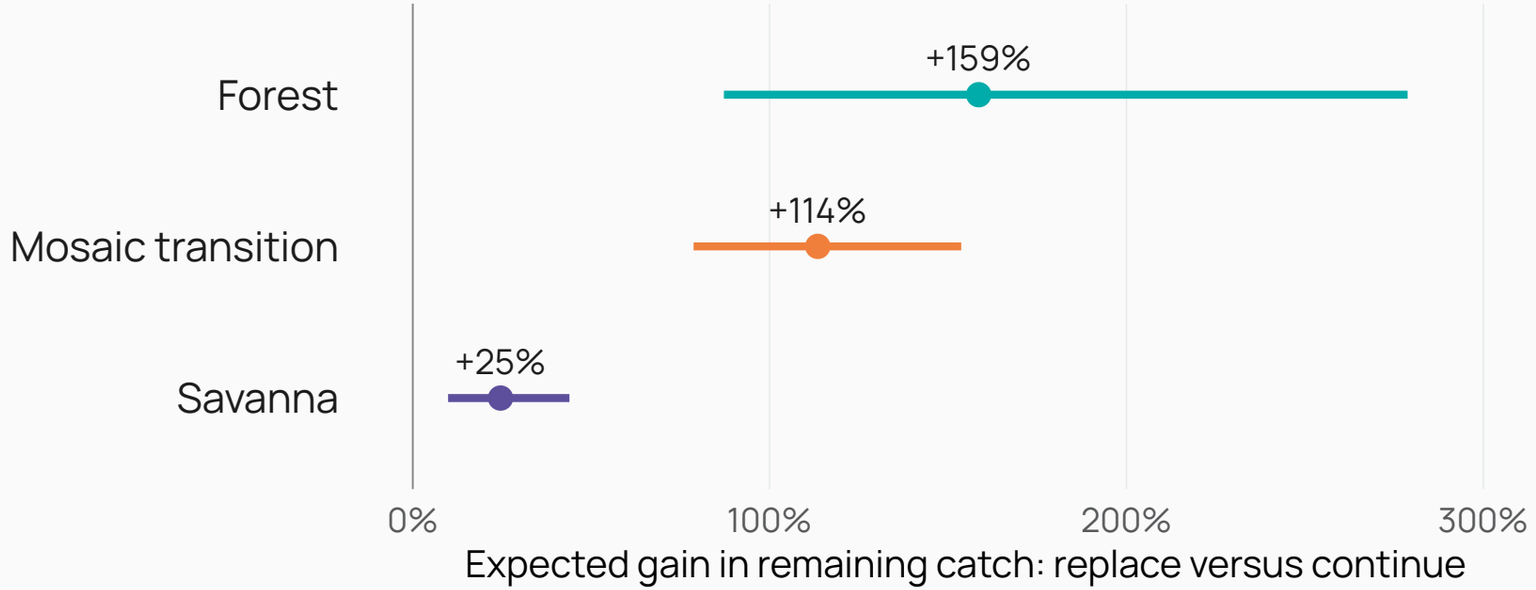
Season changes the chance of finding flies



Repeated zeros need to be interpreted in their seasonal context.

When could replacing a site improve catches?

After three zero months: compare the expected catch over the remaining nine months.



The potential gain depends on ecological setting and when sampling starts.

Modelled scenario. Dots show the mean gain across start months.
Lines show the lowest and highest gains across start months.

An adaptive strategy to test

1 **Prepare original and reserve sites**

Use the same selection process for both.

2 **Choose when to replace a site**

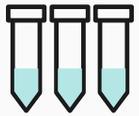
Compare **1–4 initial zero months**, accounting for season and start month.

3 **Validate with parallel field sampling**

Follow original sites for **12 months** and sample reserves alongside them.
Compare fly yields over the same calendar months.

From blackfly sampling to elimination assessment

Next research steps



Test the collected flies

Use infectivity tests to investigate where infective flies occur.



Combine human and vector evidence

Develop joint models of epidemiological and entomological observations.

Better sampling strengthens the evidence for elimination assessment.
Fly yield alone does not establish interrupted transmission.

Conclusions

- **Spatial predictions** help target sampling within accessible areas.
- **Seasonal evidence** helps teams interpret repeated zero catches.
- **Accessible web tools** put sampling plans in the hands of field teams.

Next research steps

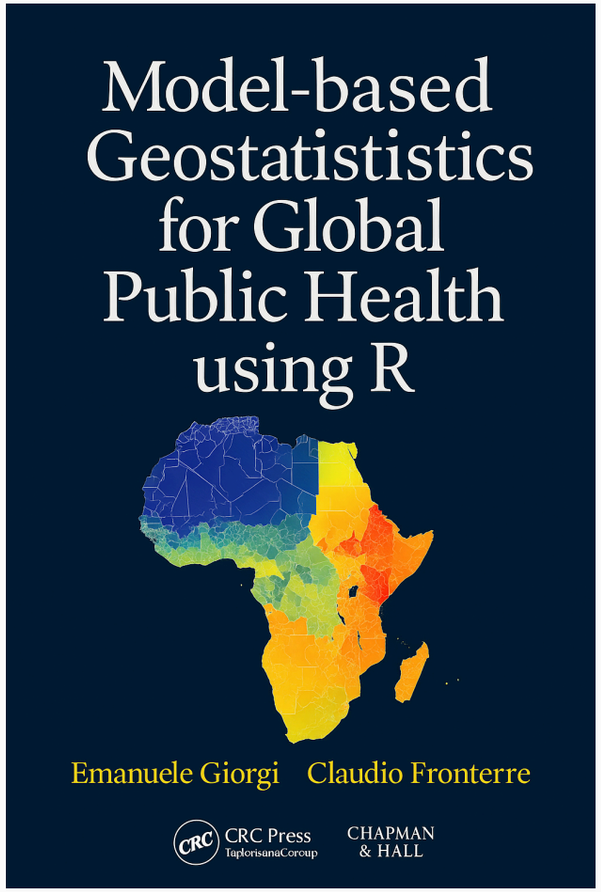
Field validation

Test the adaptive strategy through prospective comparisons of fly yield.

A joint epi–ento model

Combine epidemiological and entomological evidence to assess transmission.

Coming soon: a new book



Model-based Geostatistics for Global Public Health using R

Emanuele Giorgi and Claudio Fronterre

A hands-on introduction to disease mapping with the **RiskMap** package and real-world case studies.

Read online at **mbgr.org**.

Acknowledgments

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Entomologists and fly collectors

Communities supporting field sampling

Grazie!

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Backup: models and analysis datasets

Analysis	Dataset and purpose
Blackfly Finder	676 locations; total catch and observed duration
Observed seasonality	664 sites; 7,513 site-months
Early versus later catches	374 sites; first complete 12-month window
Adaptive scenarios	Latent zero propensity and monthly seasonal exponents

The mapped model, descriptive analyses and adaptive scenarios answer different questions.

Backup: the mapped hurdle distribution

Let Y be total fly catch at a site and $p = P(Y > 0)$.

$$P(Y = 0) = 1 - p$$

$$P(Y = y) = p \cdot f_{NB}(y; \mu, \theta) / [1 - f_{NB}(0; \mu, \theta)], \quad y \geq 1$$

$$E(Y) = p \cdot \mu / [1 - (\theta / (\theta + \mu))^{\theta}]$$

The positive-count distribution is zero-truncated. μ is the underlying negative-binomial mean, not the truncated mean.

Backup: presence score and calibration

Candidate learners: elastic-net logistic regression, XGBoost and random forest.

$$q(x) = \Sigma_j \, w_j \, q_j(x)$$

Saved weights: elastic net **0**, XGBoost **0**, random forest **1**.

$$\text{logit}\, p(x) = -3.033 + 6.016 \, q(x)$$

The mapped hurdle model applies this logistic regression to the saved prediction score.

Backup: fitted random-forest presence score

500 trees, mtry = **4**, minimum node size = **10**, Gini splitting.

```
soilT_1_0_5cm_r
ndti
CRFVOL_WeigAver_r
slope_grad_dw_cel
soilT_2_0_5cm_r
rough_scale
soilT_9_0_5cm_r
AWCtS_WeigAver_r
occurence_proximity
rough_magnitude
LC2021
SBIO4_Temperature_Seasonality_5_15cm_r
CLYPPT_WeigAver_r
SLTPPT_WeigAver_r
CHELSA_bio14_r
sti
CHELSA_bio11_r
SNDPPT_WeigAver_acc
stream_dist_dw_near
convergence
LS_Ct
ndwi
```

The saved prediction bundle retains these 22 environmental inputs and training medians for imputation.

Backup: positive-catch model

$$\log \mu = \log T + 2.173 - 0.581\, x_1 + 1.029\, x_2 - 0.273\, x_3 - 0.281\, x_4$$

Predictor	Retained input
x_1	Mean diurnal air-temperature range
x_2	Soil clay content
x_3	Soil-temperature seasonality, 5–15 cm
x_4	Terrain deviation scale

T is the number of observed sampling months. NB dispersion $\theta \approx$ **0.299**.

Backup: validation and calibration limits

- Location-level cross-validation compares candidate hurdle models.
- Candidates are ranked by RMSE, then MAE and deviance.
- Base presence predictions are out of fold; weights and upstream selection reuse outcomes.
- The reliability diagram shows an isotonic curve and a 90% pointwise consistency region.

Validation in new geographic settings and prediction of exceptional catches remain important.

Backup: empirical early–late comparison

First 3 months	Positive catch in months 4–12	No later catch
At least one fly	78 / 83 = **94.0%**	5 / 83
Zero flies	46 / 291 = **15.8%**	245 / 291

One complete consecutive 12-month period per site. Missing months are excluded.

For k = 1–4, the right-hand figure conditions on k initial zero months and observes months k+1–12.

Backup: learning from repeated zeros

Let X be a latent site probability of zero catch, with density $f_0(x)$.

$$M(r) = E(X^r) = \int_0^1 x^r \, f_0(x) \, dx$$

$$f_k(x \mid k \text{ zeros}) = x^k \, f_0(x) \, / \, M(k)$$

$$P(\text{next catch} > 0 \mid k \text{ zeros}) = 1 - M(k+1) \, / \, M(k)$$

This non-seasonal model updates the site's latent productivity after observing zeros.

Backup: seasonal adaptive scenario

For ecological zone z and calendar month t:

$$P(\text{zero in month } t \mid X, z) = X^{s_z(t)}$$

$$S_k = \Sigma_{\text{first k months}}\ s_z(t)$$

$$E(C_{stay} \mid \text{zeros}) = \nu\ \Sigma_{\text{remaining t}}\ [1 - M(S_k + s_z(t)) / M(S_k)]$$

$$E(C_{replace}) = \nu\ \Sigma_{\text{remaining t}}\ [1 - M(s_z(t))]$$

Both options use the same remaining calendar months.

Backup: gains and sensitivity

$$\text{Gain}\ (\%) = 100 \cdot [E(C_{\text{replace}}) - E(C_{\text{stay}})] / E(C_{\text{stay}})$$

- Compare initial zero runs k = 1, 2, 3 or 4.
- Evaluate every calendar start month and ecological setting.
- Consider retaining the first replacement or applying the rule again.
- A coupled extension allows positive yield to depend on zero propensity.

The displayed three-zero-month gains are scenario results, not realised field gains.

Backup: site selection and future inference

- Filter candidate sites by river, road and community access.
- Rank by predicted positivity and enforce minimum site separation.
- Check major-river representation in the initial design.
- Replacements should follow the original selection process.

The planned joint epi–ento model would combine human exposure and vector infectivity evidence. Its specification is future work.